AMMA

AI-Enabled Maternal Monitoring Assistant

Design Suggestions & Additions

For Team Discussion — YCCE Electronics Engineering (VLSI) · 2026–2027

Section 1 — Suggestions by Priority

Suggestions are ordered from highest to lowest priority. Red = must fix before building. Orange = should include. Green = good to have.

PRIORITY 1 Critical

Reassign MPU-6050 as Activity Classifier — Not a Kick Detector

Suggestion	The MPU-6050 is mounted inside a plastic pod on the belt exterior. At 0.4g threshold it detects maternal footsteps, not fetal kicks (which are only 0.05–0.15g at the skin). Reassign its role entirely: use it only to classify what the mother is doing — lying still, sitting, walking, or active. Use this state to gate the piezo sensor: only trust kick readings when the mother is still or sitting.
Impact	*Without this fix, every footstep is counted as a kick. The entire clinical value of the project collapses. This is the single most important correction.*

PRIORITY 2 Critical

Replace Piezo Disc with PVDF Film Strip

Suggestion	A rigid piezo disc (buzzer type) has poor skin conformity and a narrow frequency band. Mounted inside a plastic pod, it is separated from the skin by air and plastic, severely attenuating the fetal kick signal. Replace with a PVDF (polyvinylidene fluoride) film strip sewn directly into the inner fabric layer of the belt for full skin contact. PVDF film is flexible, broadband, and highly sensitive to low-force pressure waves.
Impact	*PVDF film is the primary sensing element of the entire project. Without it, fetal kick detection will not work reliably on a real person. Non-negotiable.*

PRIORITY 3 Critical

Add ADS1115 External ADC — Bypass ESP32 Internal ADC for Analog Sensing

Suggestion	The ESP32's internal ADC has approximately 50mV of noise and is directly interfered with by the Wi-Fi radio at the silicon level. When Wi-Fi transmits, ADC readings spike randomly. A fetal kick signal from PVDF film is only a few millivolts — this noise completely masks it. Add an ADS1115 16-bit external ADC on the I2C bus. All piezo analog readings go through ADS1115. The ESP32 internal ADC is then used only for non-critical tasks like battery voltage.
Impact	*Without clean ADC, all analog sensor data is unreliable regardless of how good the PVDF sensor is. ADS1115 reduces noise by 100x and is immune to Wi-Fi interference.*

PRIORITY 4 Important

Relocate MAX30102 to Wrist or Finger — Remove from Abdomen

Suggestion	The MAX30102 is placed in the right sensor pod on the abdomen. Abdominal skin is too thick and poorly perfused for reliable PPG. SpO2 and heart rate readings from this placement will be clinically incorrect. Move the MAX30102 to a wrist strap or finger clip connected by a thin cable to the main electronics pod.
Impact	*Incorrect SpO2 values on the dashboard are a patient safety issue if a mother or doctor acts on them. This fix makes maternal vitals clinically meaningful.*

PRIORITY 5 Important

Implement Dual-Sensor Kick Validation Rule

Suggestion	A single sensor threshold crossing is not sufficient to confirm a fetal kick. Any sharp transient — belt adjustment, maternal cough, sitting down quickly — will be counted. Require both conditions simultaneously: (1) PVDF detects a transient above threshold, AND (2) MPU confirms mother is in a low-activity state. Also add a minimum inter-kick interval of 2 seconds — real kicks cannot occur faster.
Impact	*Dramatically reduces false positives. Your detection accuracy metric in bench testing will improve significantly. This is what makes the system usable in real life.*

PRIORITY 6 Important

Replace High-Pass Filter with Bandpass Filter (2–20 Hz) on Piezo Signal

Suggestion	The original firmware uses a 1Hz high-pass filter on the piezo signal. This lets through maternal heartbeat artifacts at 1.2Hz and breathing at 0.3Hz, which will generate false kick events when the mother is resting. Replace with a bandpass filter of 2–20Hz. A fetal kick's mechanical signature falls squarely in the 3–15Hz range. Implement as a 2nd-order IIR Butterworth filter — approximately 10 lines of firmware code.
Impact	*Eliminates the two most common sources of false alerts during rest: maternal heartbeat and breathing vibration. Directly improves signal quality without any hardware changes.*

PRIORITY 7 Important

Fix Alert Engine — Use Firestore Trigger Instead of Hourly Schedule

Suggestion	The current alert engine runs on an hourly Cloud Function schedule. This means a critical movement drop at 11:05 AM may not trigger an alert until 12:00 PM — up to 55 minutes lost from the 2–3 hour intervention window your project is built around. Replace with a Firestore onCreate trigger: every time the ESP32 writes a new reading document, the alert engine runs immediately. Alert latency drops from 60 minutes to under 5 seconds.
Impact	*The 2–3 hour intervention window is your core clinical design principle. An alert latency of up to 60 minutes wastes nearly half of it. This fix is a single configuration change in Firebase.*

PRIORITY 8 Good to Have

Add 72-Hour Cold Start Strategy for Personalised Baseline

Suggestion	The AI engine computes personalised z-score baselines, but the proposal never explains what happens during the first 72 hours. If movement is low from day 1, the system learns a low baseline and never alerts. During the first 72 hours, use the standard clinical population baseline (10 kicks per 2-hour session per ACOG guidelines). After 72 hours, transition to the personal z-score model. Show on the dashboard: 'Building your personal baseline — 2 days remaining.'
Impact	*Closes a patient safety gap. Prevents the system from silently accepting abnormally low movement as normal during the critical setup period.*

PRIORITY 9 Good to Have

Upgrade Battery to 2000mAh

Suggestion	Current draw with ESP32 active and Wi-Fi on is approximately 270mA total. A 1000mAh battery gives only 3.7 hours — less than the 6–8 hour session target. Upgrade to 2000mAh. Combined with duty-cycled Wi-Fi (active for 2-second bursts every 30 seconds only), estimated runtime is 7–9 hours. Same TP4056 charger, same form factor.
Impact	*Without this, the belt dies mid-session. The 6–8 hour working mother use case — the core of your problem statement — is not achievable with a 1000mAh battery.*

PRIORITY 10 Good to Have

Move Electronics Pod to Back of Belt — PVDF in Fabric Layer

Suggestion	Rigid plastic pods on the front abdomen cause pressure points on a pregnant belly during 6–8 hour wear. Redesign with the electronics pod at the back of the belt (like a bra clasp position) and the PVDF film sewn into the inner fabric layer. This also naturally orients the ESP32 antenna away from fetal tissue, resolving the RF safety concern without any firmware configuration.
Impact	*Comfort is a clinical outcome. A belt that causes discomfort will not be worn, making all technical accuracy irrelevant. Also eliminates the RF safety concern automatically.*

Section 2 — Additional Sensors & Components to Consider

These are sensors and components not in the original proposal that would meaningfully improve the system. Each is justified by a specific clinical or technical gap.

ADS1115 — 16-bit External ADC

Why Needed	The ESP32's internal ADC is too noisy for the small analog signals produced by the PVDF film. This is not optional — it is required for the primary sensing function to work.
What It Does	Provides clean 16-bit analog-to-digital conversion on the I2C bus. Completely isolated from the ESP32's Wi-Fi radio interference. Replaces all critical analogRead() calls.
Benefit	Enables the PVDF film to actually detect fetal kicks reliably. Without this, the sensing system does not work regardless of other improvements.

MLX90614 — Infrared Skin Temperature Sensor

Why Needed	Reduced uterine blood flow — a precursor to placental insufficiency — is associated with a measurable drop in abdominal skin temperature. Your current sensor suite has no thermal parameter.
What It Does	Non-contact infrared thermometer on I2C. Placed on the inner surface of the electronics pod to read abdominal skin temperature without direct pressure. Adds a third physiological signal to the multi-parameter alert scoring alongside movement and maternal heart rate.
Benefit	Enriches the AI engine with a thermal parameter. A combination of reduced movement plus falling skin temperature is a stronger distress signal than movement alone. Connects to the existing I2C bus with no new wiring complexity.

AMS1117 3.3V Linear Regulator

Why Needed	Powering all I2C sensors (MPU-6050, ADS1115, MLX90614, MAX30102, OLED) directly from the MT3608 boost converter output introduces switching noise into sensitive sensor power rails. This noise appears directly in sensor readings.
What It Does	A dedicated 3.3V linear regulator provides a clean, noise-free power rail specifically for all sensors. The ESP32 continues to use the boost converter output. Linear regulators reject high-frequency switching noise that boost converters generate.
Benefit	Reduces sensor reading instability caused by power rail noise. Particularly important for the ADS1115, whose precision is only as good as its reference voltage quality.

Resistive Flex Sensor or FSR (Force Sensitive Resistor)

Why Needed	The system currently has no way to know whether the belt is being worn or is sitting on a table. If the belt is unworn, the system will log zero kicks and potentially trigger a false urgent alert. There is also no way to detect if the belt has shifted off position.
What It Does	A thin resistive flex sensor or FSR strip sewn into the belt detects skin contact pressure. When the belt is worn correctly, the sensor reads above a threshold. When the belt is removed or loose, it reads near zero. This data feeds into the session quality score.
Benefit	Prevents false alerts caused by an unworn belt. Enables the session quality score feature. Tells the dashboard to show 'Belt not detected — please check fit' instead of a false distress alert.

ESP32-S3 (Upgrade from Standard ESP32)	
Why Needed	The standard ESP32 uses BLE 4.2 and requires a separate USB-UART chip for programming. For a project targeting rural deployment where phone relay over BLE is essential, BLE version and range matter.
What It Does	The ESP32-S3 offers BLE 5.0 (longer range, lower power), native USB (no extra USB-UART chip), and vector processing instructions useful for running the IIR bandpass filter faster on Core 0. Firmware is approximately 95% compatible — minor pin mapping changes only.
Benefit	BLE 5.0 provides longer reliable range between the belt and the mother's phone, which is critical for the rural deployment scenario where a companion app relays data over mobile data instead of Wi-Fi.